Data and Visual Analytics

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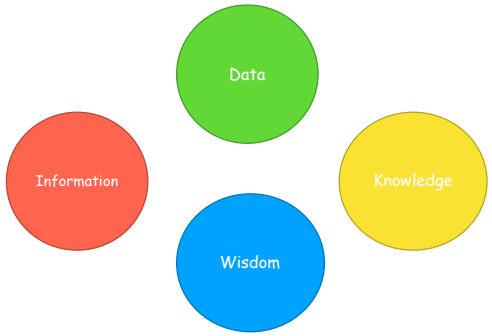


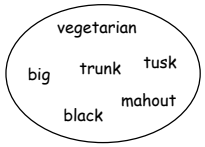
JUST START WITH THIS !

DATA

- Data is discrete, objective facts or observations, which are unorganized and unprocessed and therefore have no meaning or value because of lack of context and interpretation.
- Data is recorded (captured or stored) symbols, including , words (text and/or verbal), numbers, diagrams, and images (still &/or video), which are the building blocks of communication"

DATA, INFORMATION & KNOWLEDGE

Elephant

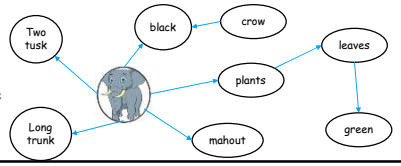


Data

- Biggest animal in the land
- Has two tusk
- Black in Colour
- Has a long trunk
- Eats plants
- Mahout is the keeper

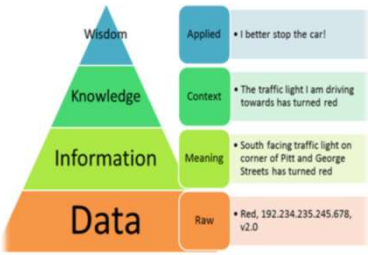
Information

Knowledge



AT GLANCE

- Data – facts and statistics collected for reference or analysis.
- Information – Facts provided or learned about somethingor someone.
- Knowledge – Facts, information, and skills acquired by a person through experience or education.
- Wisdom – the quality of having experience, knowledge, and good judgment – or my favorite…knowledge rightly applied.



WHY DATA IS IMPORTANT ?

- Make Informed Decisions
- Get The Results You Want
- Find Solutions To Problems
- Back Up Your Arguments
- Be Strategic In Your Approaches
- Keep Track Of It All
- Know What You Are Doing Well

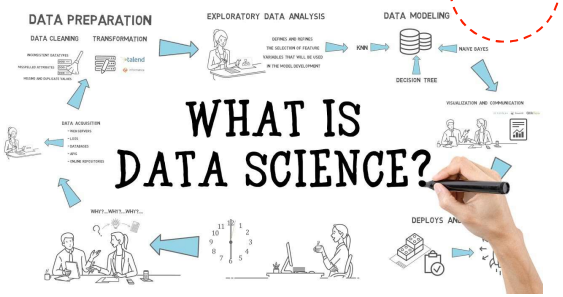
DATA IS NEW OIL

REALITY ABOUT DATA

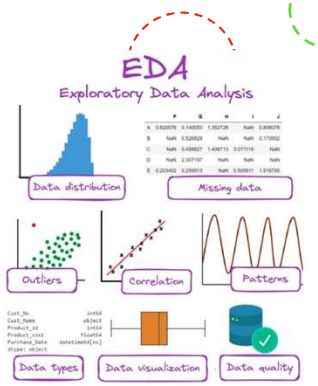
We are now **drowing** in data ! but **starving** for insights ☹

SOLUTION ?

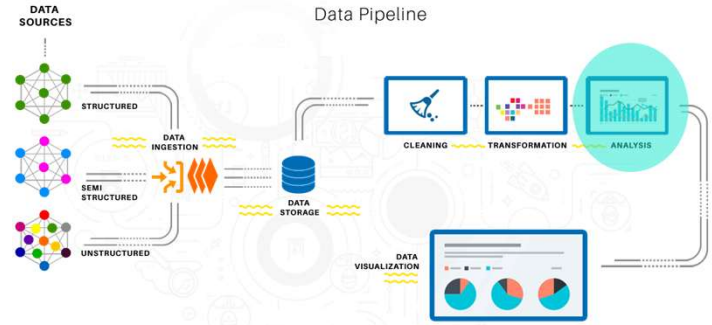
DATA SCIENCE



DATA SCIENCE



DATA SCIENCE PIPELINE



NEED AND TYPE OF DATA ANALYTICS

NEED

- Describe past results
- Diagnose why past results occurred
- Predict what might happen in the future
- Recommend actions for the organization's next steps

TYPES

- Descriptive Analytics.
- Diagnostic Analytics.
- Predictive Analytics.
- Prescriptive Analytics

DESCRIPTIVE ANALYTICS

***What* Happened ?**

- This type of analytics is rooted in facts
- Descriptive analytics can serve as that starting point or base for building your analytics strategy.
- With some professionals estimating that 80% of analytics involve descriptions based on aggregations of past performance.
- Techniques like data aggregation, data mining, clustering and/or summary statistics all serve to provide analytics that describe a past state—descriptive analytics

- A report provided by your sales team may include
  - Information on new clients, win rates and goals.
  - To customize the data view to look at only specific data to answer a specific question, such as
    - "How many customers do we have in Kochi?"
    - "How much volume did we sell last month in the Kochi City?"

DIAGNOSTIC ANALYTICS

***Why* Did it Happen ?**

- Diagnostic analytics also focus on the past.
- These types of analyses look for cause and effect to illustrate why something occurred.
- The objective is to compare past occurrences to determine causes.
- This is ofcourse a black-and-white answer, but instead provided in the context of probability, likelihood or a distributed outcome.

Diagnostic analytics can provide guidance by helping to:

- **Identify outliers**. For example, a sudden drop in sales or an explosion in website traffic that can't be explained may indicate a need for additional examination.
- **Isolate patterns**. Analysts may need to look outside the existing dataset to identify the source of the pattern. For example, a sudden drop in sales may have stemmed from the launch of a disruptor.
- **Uncover relationships**. Using more complex analytics, analysts may employ probability theory, regression analysis, or time series to isolate cause and effect relationships.

PREDICTIVE ANALYTICS

What Happens *If* ?

- Predictive analytics is the branch of the advanced analytics which is used to make predictions about unknown future events.
- Predictive analytics uses many techniques from data mining, statistics, modeling, machine learning, and artificial intelligence to analyze current data to make predictions about future.

- Organizations are turning to predictive analytics to help solve difficult problems and uncover new opportunities. Common uses include:
- **Detecting fraud**. Combining multiple analytics methods can improve pattern detection and prevent criminal behavior.
- **Optimizing marketing campaigns**. Predictive analytics are used to determine customer responses or purchases, as well as promote cross-sell opportunities.

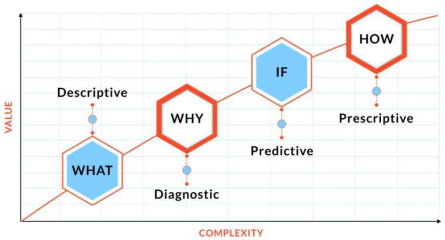
PRESPECTIVE ANALYTICS

How to Make it Happen ?

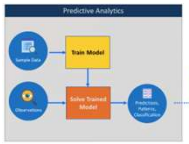
- **Prescriptive analytics** builds on predictive analytics by helping determine recommended (prescribed) actions based on desired potential (predicted) outcomes, helping organizations achieve their business objectives.
- Prescriptive analytics models are constantly "learning" through feedback mechanisms to continuously analyze action and event relationships and recommend the optimal solution.
- By simulating the solution, prescriptive analytics can examine all the key performance criteria to ensure the outcome would achieve the correct metric goals before anything is implemented.

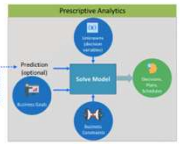
- Artificial intelligence, machine learning and neural network algorithms are often employed to support prescriptive analytics
- To make specific suggestions based on nuanced patterns and perceptions of organizational goals, limitations and influencing factors.

INDUCING INTELLIGENCE IN ANALYTICS



DIAGNOSTIC





LESS — HIGH

Need of AI

TOOLS FOR DATA ANALYTICS

BIGDATA

PSPP, Excel, Google Sheets, tableau, Power BI, Spark, HBASE, hadoop, Solr, kafka, elastic

DATA ANALYSIS
+
DATA VISUALIZATION
+
DASHBOARD DEVELOPMENT

stata, sas, python, Data Studio, Minitab, Qlik Sense, SPSS, R, Weka, orange, Java, Scala, SQL Server, MySQL, Google BigQuery, PostgreSQL

AI/ML

jupyter, SciPy, SPYDER, NumPy, DASK, bokeh, learn, pandas, TensorFlow, Numba, Panel, matplotlib, HoloViews, PYTORCH, CONDA

THANK YOU!

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QUESTIONS